PEST MANAGEMENT DECISION GUIDE: GREEN AND YELLOW LIST

Early blight fungal disease in tomato

Alternaria solani



Early blight symptoms of large light brown areas on leaves, first appear on older leaves near the plant base (photo: cornell.edu)



Dark concentric spots caused by early blight (photo Joey Williamson, HGIC, Clemson Extension)

Prevention

- Use disease-tolerant cultivars such as Nemo-nadin
- Rotate tomatoes only with non Solanaceous crops especially not with potato or okra because they also host the disease
- Plant tomato, particularly seedlings, in raised beds to improve drainage and prevent disease.
- Give tomato plants extra space if you know blight problems occur in your region (more than 60 cm) to allow proper aeration among leaves.
- Water the soil around plants and not over the plant to prevent splashing and disease spread. Do not irrigate overhead, consider drip or furrow irrigation.
- Remove and burn affected plants at the end of the season
- Mulch with black plastic or with grass to prevent fungus from spreading to leaves.

Monitoring

- Look for dark concentric spots on the lower leaves and stems (tell-tale rings).
- Lower leaves may turn yellow-brownish and drop.
- Fruits can be affected, the spots often begin near the stem of the fruit.
- The symptoms should be checked from vegetative stage to fruiting stage of tomato.
- Check for symptoms every week, and consider applying direct control measures as soon as the disease appears
- The disease is most evident during the rainy season
- The disease often breaks out after a period of heavy rainfall.

Direct Control

- Prevention is the most powerful measure. Direct control is difficult and often too late
- Remove and destroy infected leaves, but do not carry them openly through the field
- Neem tree leaf extracts can reduce fungal growth and spore formation. It prevents the fungus from colonizing the leaves. Soak pounded 1kg neem tree leaves in 5 ltrs of water for 24 hrs and use the fluid to spray after mixing it with water in a 2:1 ratio. A sticker, such as soap should also be added to the mixture.

Direct Control / Restrictions

Direct Control	Restrictions
• Prevention is the most powerful and often only measure, and chemical sprays are often too late or have little effect. The following products are mainly protectants and so may prevent further spread but cannot or only moderately suppress already established fungi.	
• When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, pre-harvest interval, max number of sprays, restricted re-entry interval. Do not empty into drains.	
• WHO class II pesticides might not be allowed in local IPM schemes.	
• Always consult recent list of registered pesticides (ZEMA).	
• Chlorothalonil -based products (Daconil and many others). Multi-side action pesticide.	• WHO toxicity class U (unlikely to present acute hazard in normal use); Pre-harvest interval (p.h.i.) 7-14 d depending on product; max 3 sprays with 10 d min. treatment interval, restricted re-entry interval (r.e.i.) 3 d; very toxic to aquatic organisms
• Mancozeb -based products. Broad spectrum contact fungicide with some protective action. Multi-side action pesticide.	• WHO class U (unlikely to present acute hazard in normal use); max 2 post planting applications, p.h.i. 14 d; r.e.i. 3 d; Avoid treating wet crops and those suffering from drought
• Copper -based product sprays (copper salts)	• For dosages, p.h.i., r.e.i. and max number of sprays refer to manufacturer's instructions

ZARI Zambia Agriculture Research Institute

Researching Soils, Crops and Water in Zambia

Zambia

CREATED/UPDATED: July 2014
AUTHOR(S): Francisca K.C. Mwanda kankumwanda@yahoo.com, Hildah Chimutingiza chimutingiza@yahoo.com, Jones Malama jozmalama@gmail.com (Zambia Agricultural Research Institute ZARI of Ministry of Agic. & Livestock, Mt. Makulu, Chilanga, Zambia), George Silumbwe (BEC, DoA, MAL), Malambo Mitcho (BEC, DoA, MAL)
EDITED BY: Plantwise

LOSE LESS, FEED MORE
PlantwisePlus is a CABI-led global initiative www.plantwise.org
©CAB International. Published under a CC BY-NC-SA 4.0 licence